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https://doi.org/10.1071/CH25175

AUSTRALIAN JOURNAL OF CHEMISTRY

openGDE/bayadjamara: an open-source 3D-printable gas diffusion electrochemical reactor platform

Christopher Barnett[A,B,C,*], Alexander K. L. Yuen[A,B], Anthony F. Masters[A] and Thomas Maschmeyer[A,B,C,*]

For full list of author affiliations and declarations see end of paper

***Correspondence to:**
Christopher Barnett
The Laboratory of Advanced Catalysis for Sustainability, The University of Sydney, Sydney, NSW, Australia
Email: christopher.barnett@sydney.edu.au

Thomas Maschmeyer
The Laboratory of Advanced Catalysis for Sustainability, The University of Sydney, Sydney, NSW, Australia
Email: thomas.maschmeyer@sydney.edu.au

Handling Editor:
Paul Bernhardt

Received: 5 November 2025
Accepted: 19 January 2026
Published: 24 February 2026

Cite this: Barnett C *et al.* (2026) *openGDE/bayadjamara*: an open-source 3D-printable gas diffusion electrochemical reactor platform. *Australian Journal of Chemistry* **79**, CH25175. doi:10.1071/CH25175

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ABSTRACT

We introduce a flexible reactor platform for testing electrocatalysts, especially for gas-phase reactants or products. This open-source, 3D-printable platform has been designed for gaseous reagents in general and is readily modified to suit a range of applications, configurations and reactor sizes. We demonstrate the versatility of our platform with iridium(IV) oxide as a standard catalyst for water splitting, using a range of conditions that relate the system's performance to the literature. To encourage adoption of our system, we have included detailed build instructions, 3D-printing tips, example reactor configurations and a guide on modifying the provided design code. It is our hope that the simplicity, low-cost and provision of design files will encourage use of the system to more easily enable appropriate testing of gas-diffusion electrocatalysts that will allow for ready cross-laboratory comparison and benchmarking.

Keywords: 3D printing, additive manufacturing, electrocatalysis, electrochemistry, gas diffusion electrode, labware, reactor design.

Introduction

Reliable data is the cornerstone of good science. The performance of different gas diffusion electrode (GDE) catalysts must be compared using data generated from an appropriate system.[1,2] This requires well-defined electrodes and precise reactor geometry that can be easily reproduced, and most importantly, a gas diffusion setup. Despite general agreement on these requirements, the performance of catalysts proposed for use in a GDE is still mostly evaluated using standard aqueous three-electrode setups,[3–5] with the gas of choice simply bubbled through the solution. There are three significant drawbacks with this approach: first, this is not a gas diffusion electrode, second, most gases have a low solubility in solution (see Solubility of Gases in Water vs. Temperature at https://www.engineeringtoolbox.com/gases-solubility-water-d_1148.html), third, the observation that often catalysts behave differently when operating in gas-in-solution systems versus pure gas or reactant-in-gas systems.[1,2,6,7]

GDEs are becoming increasingly popular research platforms, with applications in: reverse fuel cells,[6,8–13] waste remediation[14] and electrolysis,[13] as well as any electrochemical reaction where a reagent or product is a gas or is contained in a gas stream, such as CH_4,[15] CO_2[16–18] and CO.[6]

It has been firmly established that testing of GDE catalysts in a three-electrode setup can significantly skew the observed performance.[6,7] The performance estimates can be skewed either too low, thereby leading to the premature exclusion of promising catalyst candidates, or too high, thereby encouraging unprofitable further research.[1,19]

A case in point is that of the translation of laboratory results into commercial hydrogen fuel cells.[20] Even when restricting the scope to just that of platinum-based oxygen reduction reaction (ORR) catalysts, markedly different performance in liquid half-cells as compared to commercially relevant GDE-based fuel cells[21,22] has been reported for e.g. Pt-alloys,[23] de-alloyed Pt–Ni nanoparticles[19,24] and Pt–Ni nanoframes.[25,26] To address

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these inconsistencies, several calls have been made for the standardisation of candidate catalyst testing within GDE or membrane electrode assembly (MEA) setups.[6,21,22]

Although commercial GDE test reactors are available, they are expensive, non-standardised and somewhat limited in scope or adaptability. By contrast, a liquid-based three-electrode setup requires nothing more than a beaker as the reactor; though testing should ideally be performed with the working electrode (WE) and counter electrode (CE) separated, such as in U-tube cells[4,27,28] and H-cells.[29] With the rise in additive manufacturing, there has been an increase in the number of reports of 3D-printed GDE reactors.[1,30,31] However, these publications often provide insufficient detail to reproduce the reactor, and we are unaware of any that provide access to the design files.

Notably, there is no agreed upon standard GDE reactor. Instead, the common practice is to compare the performance of a new catalyst against that of a 'standard' or benchmark catalyst, within the same reactor.[32] This practice is problematic when no standard catalyst exists for a reaction, such as carbon dioxide reduction,[33] nitrogen reduction[34] and the dehydrogenation of ethylbenzene.[35] This practice also relies on reproducible benchmark catalysts to be accessible to all researchers, and for the reactor design to not mask the catalyst performance due to mass diffusion limitations or temperature gradients.[36]

It has been elegantly shown by Ehelebe *et al.* that the specific GDE reactor set-up (i.e. size, shape, orientation of the gas flow field) is less important for reliable catalyst testing than reproducible sample preparation (e.g. ensuring homogeneity and representative of the bulk) and the electrochemical protocols (e.g. not diffusion limited, appropriately referenced) used.[37] However, these protocols rely on having a means to establish, for example, the electrochemically active surface area; a problem for newly synthesised materials or unestablished reactions. Most importantly, access to a GDE/MEA test reactor is a pre-requisite for analysis of gas-phase electrochemistry. In these circumstances, access to a standardised reactor would allow for meaningful comparisons of catalyst performances.

To address these issues, we have developed an accessible, affordable and adaptable 3D-printed GDE/MEA test reactor: '*openGDE*' or '*bayadjamara*'.[A] We provide the design files and detailed instructions on how to modify and print the reactor platform, suitable for even a relatively novice operator. 3D-printing offers adaptable design, access to unique geometries not available through traditional subtractive manufacturing, and importantly, easy fabrication of multiple reactors, which drastically lowers the cost of GDE/MEA test reactors and allows for parallel testing.

For reactions with established catalysts or electrochemical protocols,[37] we are confident that *openGDE* will facilitate comparison with existing data. This research platform is designed to encourage appropriate testing of GDE/MEA catalysts by the user community to enable and enhance research quality and outcomes, i.e. testing and comparisons of catalysts under near identical and well-specified conditions – analogous to the improvements in the field of zeolite catalysis that have arisen from the introduction of a standard, verified zeolite preparation and characterisation methods.[38]

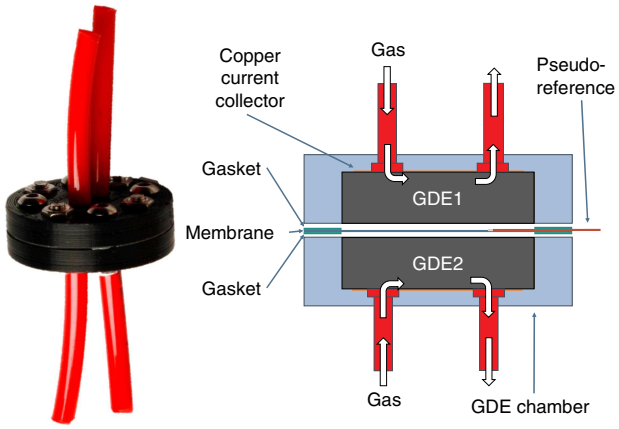

Fig. 1. Left: image of example GDE|GDE reactor, using default settings (e.g. GDE 1 cm^2). Right: simplified schematic of the reactor design.

To achieve this end, we set out to:

1. improve the accessibility of GDE reactors for research;
2. design the GDE reactor in a way that is easily adaptable to fit a variety of needs;
3. develop a platform for 3D-printable reactors, rather than a one-off design;
4. to make printing possible on even hobbyist-grade 3D printers with readily available filaments; and
5. to simplify the design and printing process to enable confident comparisons of GDE catalysts between different laboratories.

Two configurations of our 3D-printable GDE reactor for catalyst evaluation are shown in Fig. 1 and 2, with the adaptable source files being released open-source. Both the model files (.STL and .3MF) of the components used in this work, as well as the design file (.SCAD) are available on both Github and *Zenodo*, which also includes instructions on how to use the 'Customizer' feature to modify the reactor to the required parameters (see 'Component variables' in the Supplementary material). Additionally, a list of common 3D-printing terminology is provided in the Table S5 of the Supplementary material.

[A]Bayadjamara means 'air maker' in Gadi language and has been suggested as an appropriate name by Prof. Jakelin Troy, Director, Indigenous Research, Professor of Anthropology at The University of Sydney.

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Experimental

Detailed printing instructions, including tips and tricks for achieving best print results, are included in the '3D printing instructions', '3D-printing Workflow' and 'Specific instructions for the main components' sections in the Supplementary material. The reactors used for this work were printed using the settings summarised in Table 1. The printing instructions included in the 'GDE' and 'Aq' setions in the Supplementary material were followed by three other researchers, who each successfully assembled *openGDE*. Five different printers (see 'Printers tested' in the Supplementary material) were also used to print operational reactors.

Bolt tightness was tested using a Vevor torque screwdriver, with the inch–pound scale converted to Newton metres. The maximum practical pressure achievable was tested using a variation of the GDE|GDE setup of *openGDE*. Resin held up to 4 barg, and for fused deposition modelling (FDM) between 2.5 barg (untreated) and 3.5 barg (vapour smoothed), with additional details available in the 'Good seals' and 'Gasket' sections in the Supplementary material.

To ensure consistency, multiples of each iteration of a design were printed at the one time with the same settings and filament source for each component. A bed of 16 GDE chambers on conservative print speeds is easily achieved in a single 9-h (overnight) print. The design files include the option to add a label or identifier, placed in areas that minimally affect performance, to aid in the tracking of which components were used within a given experiment.

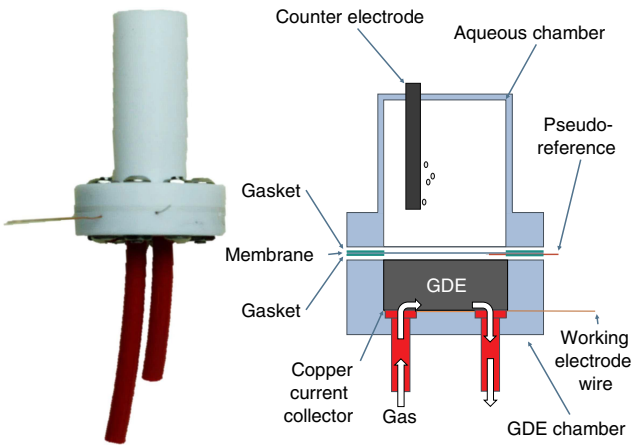

Fig. 2. Left: image of an example GDE|Aq reactor, using default settings (e.g. GDE 1 cm^2). Right: simplified schematic of the reactor design.

Discussion

In 2008, the first self-replicating 3D printer, the RepRap, produced its first child printer.[39] This achievement, as well

Table 1. Common filament comparison.

Filament	ABS	HIPS	PETG	PLA	PP	TPE	TPU	SLA
Cost	$$	$	$$	$	$$$	$$$	$$	$
Rigidity	HS	MS	HS	HS	SF	VF	F	V
Elasticity	LE	ME	M-HE	M-HE	HE	HE	HE	V
Bowden	Y	Y	Y	Y	OK	D	D	NA
H_2SO_4	E	E	E	E	E	P	E	E
KOH	E	E	P	E	E	P	P	E
Na_2SO_4	E	E	E	E	E	E	E	E
H_2O	E	E	E	E	E	E	E	E
Nozzle (°C)	190–210	230–245	235–245	210–225	235–245	235–245	228	NA
Bed (°C)	90	70	70	50	60	NA	NA	NA[A]
Bed material	PEI/Glass[B]	PEI	Glass	Glass	PP tape	Glass	Glass	NA
Chamber	Yes	Yes	Yes	Opt.	Yes	Opt.	Opt.	Yes

Cost refers to the relative amount per kilogram from online retailers in Australia. Rigidity and Elasticity are: H (high), M (medium), L (low), S (stiff), F (flexible), E (elastic), V (varies with material. For example, SLA resins are sold with varying characteristics). The Bowden measure is whether the material is suitable to print with a Bowden extruder setup. Y (yes), OK (depends on tuning), D (difficult, not recommended for beginners). H_2SO_4, KOH, Na_2SO_4 and H_2O concentrations were 0.1 M; E (excellent), P (poor) refer to how compatible the filament is with the listed solution. Bed is the recommended bed temperature during printing. Bed material is the recommended bed material. PEI (polyetherimide) and garolite are common bed materials available for purchase. Most packaging tapes are made of PP (polypropylene). Recommendations are limited to simple bed materials, not engineering grade. The Chamber measure is whether a chamber or enclosure is required for suitable prints. Opt. is optional, though a chamber is always recommended to contain the volatile components that are produced when printing. ABS, acrylonitrile butadiene styrene; HIPS, high-impact polystyrene; PETG, polyethylene terephthalate glycol; PLA, polylactic acid; TPE, thermoplastic elastomers; TPU, thermoplastic polyurethane; SLA, stereolithography.

[A]SLA/resin prints can fail at low ambient temperatures.

[B]A thin layer of ABS dissolved in acetone is applied to the surface and allowed to dry before printing.

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as the release of the printer designs under the free and open-source MIT license, started the additive manufacturing boom that has seen the release of hundreds of commercially available 3D printers – from DIY kits to fully assembled turn-key solutions. Over the years, through a combination of improved hardware, as well as advanced (including AI-enhanced) software, the quality of printing has improved to the point of commercial products, including 3D-printed components. 3D-printing has become available to the masses, with the barrier to entry low enough that many printers are available that are plug-and-play and affordable (prices as of 6 January 2026), for example:

- US$179, Creality Ender-3 V3 SE (see https://store.creality.com/products/ender-3-v3-se-3d-printer).
- US$239, ELEGOO Neptune 4 pro (see https://us.elegoo.com/products/elegoo-neptune-4-pro-fdm-3d-printer)
- US$169, Anycubic Kobra 3 (see https://store.anycubic.com/products/anycubic-kobra-3).

3D-printing has evolved beyond that of a tinkerer's hobby – where maintenance and modification of the printer was the goal. Many 3D printers are now available as a 'tool', rather than a 'project'. These printers are reliable and do not require intimate knowledge of how the machines work to achieve reliable prints. It is our hope that *openGDE* comes to fill the same niche, allowing researchers to 3D-print a reactor without having to delve deeply into computer aided design (CAD), while providing an open-source community to advance the platform.

Materials choice

Along with the advances in accessibility, the available materials for printing are becoming ever more numerous. Filaments are available with additives to control the printing and physical properties, as well as optional colourants added for surface finish and artistic purposes. In the context of *openGDE*, chemical compatibility with the solvents and reagents used is paramount. There is also the potential of contaminants in the filaments, such as metal particles,[40] or indeed constituents of the filament itself.

The final consideration is ease of printing. Although polyether ether ketone (PEEK) and polyether ketone ketone (PEKK) are both highly chemically resistant thermoplastics, they cannot be printed on a hobby 3D printer; as this requires advanced hot-ends that reach temperatures well above those achievable with such systems, and need an enclosed and actively heated chamber to avoid catastrophic warping and de-lamination. Even with these requirements met, the geometries and surface finishes achievable make it challenging to get a water-tight reactor with PEEK and PEKK.

The physical properties and chemical compatibility of several commercially available filament materials were tested for suitability for use in *openGDE*. The testing methodologies and details are available in the 'Experimental details' section of the Supplementary material, and the results are summarised in Table 1. PLA is recommended as the most suitable filament for the majority of applications due to its broad chemical resistance and ease of printing on most machines.

Dimensions and configurations of *openGDE*

openGDE is designed to be a testing platform, adaptable to many needs and requirements. For our experiments, we selected a GDE disc diameter of 11.2 mm, as this provides a membrane interface of 1.00 cm^2, simplifying calculations. For this size of disc, eight M3 bolts were ideal. If scaling up, larger bolts would provide a better seal; this is available in the customiser as a single bolt size variable. See the 'Component variables' section in the Supplementary material for more details on customising the *openGDE*, with examples.

There are numerous excellent papers that detail reactor configurations for various applications,[32,41,42] and an in-depth discussion is beyond the scope of this paper. As well as adaptable dimensions, *openGDE* includes options for different configurations. Two thoroughly tested variants are discussed below.

A commonly reported setup is GDE|GDE, shown in Fig. 1, where both reactions are in the gas phase.[1,32,43] A MEA describes a membrane coated with a different catalyst on each side, and two gas flow fields.[32] This MEA configuration is included in *openGDE* by use of a flow field insert printed from conductive filament, or a copper disc for edge contact, *vide infra*.

However, for many GDE catalyst performance measurements, there is no need for the counter reaction to also be in the gas phase. A full GDE|GDE set-up represents an unnecessary complication when comparing GDE catalysts.[2,36,37,44] We therefore designed an aqueous chamber for the counter reaction (Fig. 2). The authors are unaware of a commercial reactor that includes this set-up, could only find two examples of this concept within the literature,[2,36,37] and note that no design files are provided to accompany those examples.

The *openGDE* aqueous chamber (Aq) is designed to provide simple assembly, with adequate access to the nuts and bolts, and the default sizing also allows for sufficient solution volume (3 mL) and membrane surface area (1 cm^2). This setup provides reliable performance evaluation for the GDE catalyst,[2,36] without the included complexity of managing two gas streams, particularly when the counter reaction has diffusion limitations or sluggish kinetics.[37] Complete instructions and guideline sizing limits are available in the 'Some design choices/3D printing tricks' and 'Component variables' sections of the Supplementary material.

Components of an electrochemical cell

The various methods to integrate the essential elements of an electrochemical cell into *openGDE* are briefly discussed here.

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Gas flow

The default version of *openGDE* uses 4-mm outer diameter nylon tubing as the inlet and outlet. It was found most convenient to use a short section of tubing (5 cm) and connect to longer tubing as needed with pneumatic fittings. The tube end was heat-moulded, either against glass or a resin printed mould (see the 'T-shaped tubes' section in the Supplementary material for additional details).

Electrical

An electrical connection between the catalyst and the outside of the reactor was achieved by using an enamelled copper wire inserted through the reactor wall. This wire was then connected to an optional copper disc as current collector (see Fig. 3); removing the copper disc did not significantly affect the performance. Wire with a diameter of 0.2 mm was used, which has a current rating up to 200 mA, meaning that it is more than adequate for the testing included here (<10 mA).

Reference

Reference electrodes in a MEA or GDE setup are non-trivial. For our testing, we successfully implemented three set-ups:

1. The reactor can be run in symmetric mode, where the counter electrode is the reference electrode.
2. A pseudo-reference electrode is possible through insertion of either of a platinum wire or a silver-chloride-coated silver wire on the working electrode side of the membrane.
3. If the reactor is configured to include a liquid chamber, a standard reference (such as Ag/AgCl or Hg/HgO) can be used.

Membrane

Only commercial membranes have been tested in *openGDE*. They were sealed with the 3D-printed gasket, and the additional support provided by the bolts holds them in place.

Seals

The seals are very important to a functional reactor. Smooth mating surfaces and dimensionally matched gaskets are key for adequate sealing.

GDE/MEA

Depending on the configuration required, two methods of supporting the catalyst have been included here. The first uses a thick (0.5–2.0 mm) self-supporting carbon layer disc. The second, based on a thin gas diffusion layer (gdl), adds a flow field plate into the GDE half-reactor. The latter version is more complex, with either conductive filament or copper plating recommended for the flow field plate.

Aqueous

When using an aqueous chamber, it was found that prolonged exposure of one side of the membrane to an aqueous solution and to a comparatively dry gas on its other side could result in deformation of the membrane towards the aqueous side, eventually leading to a physical separation from the electrical contact with the GDE catalyst. A mesh

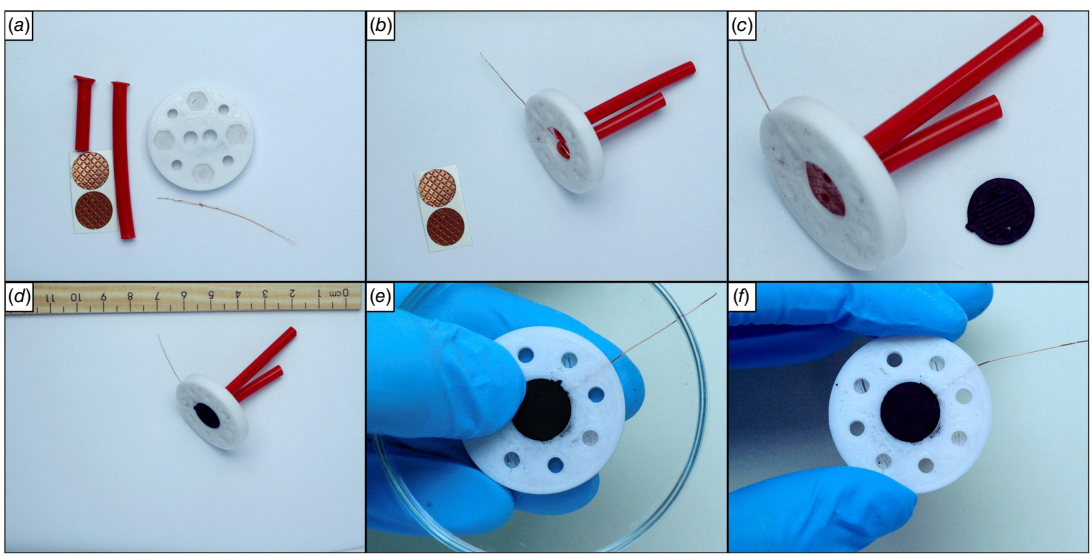

Fig. 3. Example assembly of a GDE sub-assembly, both the MEA and GDE versions. (*a*) Components needed: two T-shaped tubes, enamelled copper wire, 3D-printed GDE component (white PLA), and optional laser-cut copper tape. Not shown: gas diffusion layer (gdl) flow field insert (*c*, *d*) or GDE disc (*e*, *f*). (*b*) Gas tubing and enamelled wire installed. (*c*) Copper tape installed. Parallel flow field gdl insert shown next to component. (*d*) Assembled MEA half cell, with parallel gdl insert. Note the alignment pin. (*e*) A glass Petri dish being used to evenly insert a GDE disc into the reactor. (*f*) Assembled GDE sub-assembly.

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and a spacer insert were designed to hold the membrane in good contact with the GDE.

Assembly overview

The aim of *openGDE* is to provide an affordable and easy to construct testing platform. We therefore give a brief overview of the components needed (Fig. 4) and assembly steps involved with constructing *openGDE*. The Supplementary material includes more complete assembly instructions ('Reactor assembly guides' section), a cost breakdown (Table S3), and an example video of a reactor assembly (see CH25175_VideoS1.mkv).

The basic steps to assemble a GDE|Aq reactor are:

1. Gather M3 hardware and print the required half-reactors plus gaskets. Prepare the catalyst discs and activate chosen membrane.
2. Assemble GDE sub-assembly (Fig. 3). Insert tubing and electrical wires.
3. Prepare the gasket-membrane-gasket sandwich and pierce holes.
4. Insert catalyst disc and screw reactor together. Tighten just before use.

Electrochemistry

The capabilities of *openGDE* as a suitable reactor for a range of electrochemical conditions are demonstrated in Fig. 5. The experimentally simpler GDE|Aq reactor configuration was employed and printed using readily available PLA filament.

As mentioned above, there are three methods to include a reference in *openGDE*: (i) symmetric-mode, (ii) insertion of a pseudo-reference or (iii) use of a standard reference electrode (e.g. Ag/AgCl) in the counter reaction aqueous chamber. Each version has its advantages and disadvantages. Running a reactor in symmetric mode is commonly reported.[32,45,46] However, this requires a standardisation of the reactor using a benchmark catalyst to provide insight into the true over-potential of the reaction occurring at the working electrode. The potential of the counter reaction can vary over the course of a reaction, and the iR losses should be compensated to account for the membrane between the electrodes.[37] The success of the pseudo-reference method depends on the nature of the reaction taking place. Care needs to be taken to ensure that the pseudo-reference is not participating in the reaction and is not becoming fouled during the course of the experiment.[44] Finally, the use of a standard liquid reference is only possible when the counter reaction is in the liquid (using the Aq chamber); however, the iR losses still need to be considered.[36,37]

The following dataset was collected in symmetric mode and using a known electrocatalyst (IrO_2) to act as benchmark. Despite the limitations mentioned above, this is currently the literature standard for GDE reactors.[32,45,46] As shown in Fig. 5, the IrO_2-free GDE performed worse for the oxygen evolution reaction (OER) and better for the hydrogen evolution reaction (HER), as expected.[45,46] However, caution should be taken when comparing data generated in this fashion, as the reference electrode, when connected to the counter electrode, must change in potential as current passes. Further discussion and sample data are available in the 'Experimental details' and 'Future Work' sections and Table S5 of the Supplementary material.

Most importantly, this method of benchmarking is *only* possible when a benchmark catalyst is available. *openGDE* enables two methods for the incorporation of a separate

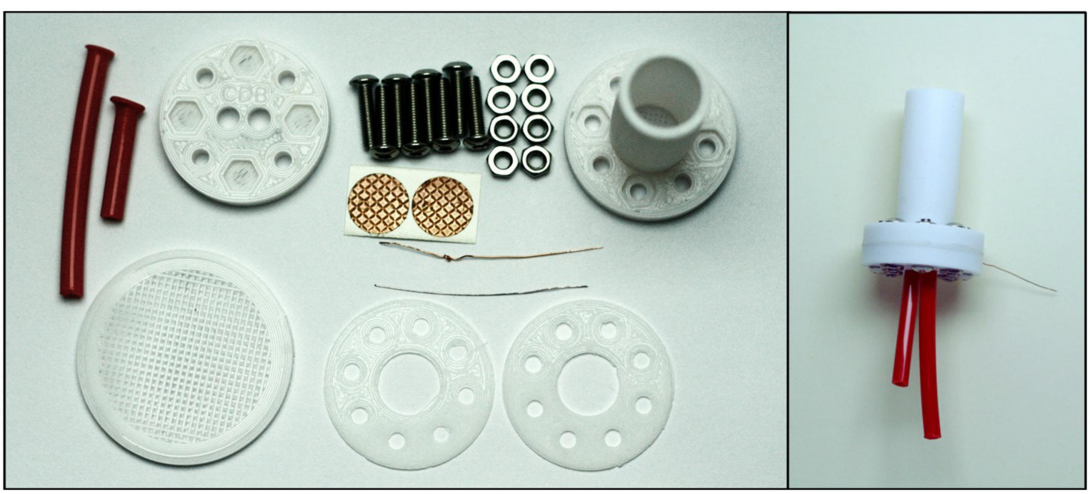

Fig. 4. Left: expanded view of components in a simple GDE|Aq reactor setup. Top left-to-right: Two T-shaped tubes, GDE component, eight M3 × 12 bolts, eight M3 nuts, Aq component. Middle: two copper tape discs, one enamel copper wire with knot, one Ag/AgCl wire as reference electrode. Bottom left-to-right: Membrane holder, two TPE gaskets. Right: assembled GDE|Aq reactor.

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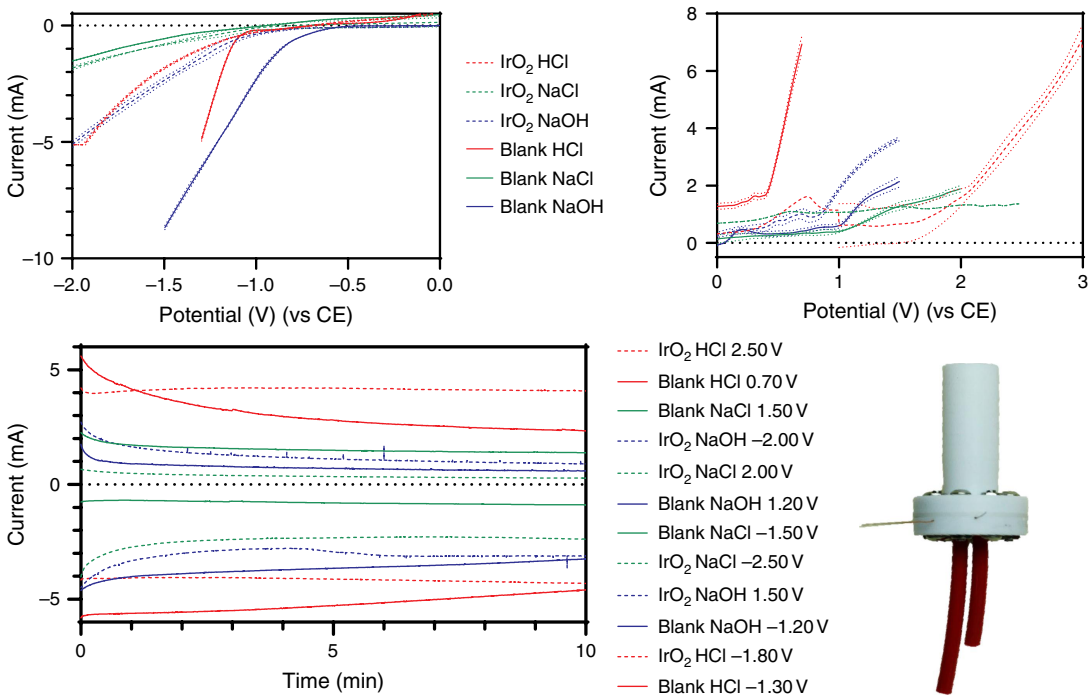

Fig. 5. Example data using the GDE|Aq configuration of *openGDE*, demonstrating its suitability for a variety of aqueous electrolytes (red, 1 M of HCl; green, 1 M of NaCl, blue, 1 M of NaOH). All data were obtained using humidified nitrogen gas flowing through a 1-cm^2 GDE WE with either no additive (solid lines) or 5 wt% commercial IrO_2 (dashed lines), the reference electrode (RE) = counter electrode (CE) = Pt wire in the aqueous electrolyte, and a commercial proton exchange membrane separating the two phases. Top: LSVs at 5 mV s^{-1} in duplicate for the various GDEs and electrolytes, average shown with standard deviation; left: negative potentials, hydrogen evolution reaction (HER) occurring in the working electrode (WE); right: positive potentials, oxygen evolution reaction (OER) occurring in the WE. Bottom: *I*–*t* curves at potentials as labelled.

reference electrode (pseudo- or Aq), while providing researchers with a readily accessible reactor of known geometry. This should allow suitable comparisons of electrocatalysts to be made without requiring a benchmark catalyst, particularly important when none is available.

Beyond simple water splitting, these data demonstrate that *openGDE* is suitable for a range of aqueous electrolytes, from highly acidic (1 M of HCl, pH = 0), to neutral (1 M of NaCl, pH = 7), to basic (1 M of NaOH, pH = 14). The reactors did not crack or leak during testing, and the self-supported catalyst discs remained intact. To test the durability, a GDE|GDE reactor made of PLA was used continuously for over 3 months without cracking or leaking. More details are available in the 'Continuous operation tests' section in the Supplementary material.

Conclusions

We have introduced *openGDE*, a design platform that is modifiable for different setups – including a gas diffusion electrode (GDE), gas diffusion layer and aqueous chamber – and demonstrated that it can produce data comparable to, or better than, reported commercial reactors.

openGDE could represent the framework for a 'standard' reactor, i.e. a widely accessible reactor allowing all interested researchers access to a reliable and reproducible comparison of GDE, gdl/MEA and related catalysts. This is particularly relevant when no benchmark catalyst is available.

To our knowledge, this is the first reported 3D-printed reactor platform that provides all source files on Github and *Zenodo* (doi:10.5281/zenodo.18322571). These are provided for use and modification under an open-source license to encourage their use and adoption by the research community, no matter their previous electrochemical or 3D-printing experience.

Flexibility and ease of modification, in the form of dimensions and configurations, were considered when designing the reactor platform to enable researchers to use the code without having to learn how to code themselves. As well as limiting parameters to sensible values, many print and material parameters have been tested, and best practices determined, to provide guidance on achieving reproducible results. We encourage all potential users to consider what is required for their particular research and to innovate and share accordingly.

The reactor has been tested without leaks in resin up to 4 barg and with extrusion printing between 2.5 barg (untreated) and 3.5 barg (vapour smoothed). Long-term

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stability testing at electrochemically relevant conditions showed the reactor capable of continuous operation for over 90 days.

openGDE is designed to be affordable and reproducible. The cost of all parts required (even when including the printer itself and consumables) is lower than one commercial reactor of equivalent size. A generally useful benchmarking-sized reactor, consisting of a GDE|GDE setup with 1-cm^2 catalyst surface area, costs less than A$2 in materials to produce. We invite researchers to contact us to organise an exploratory kit provided at cost.

Supplementary material

Supplementary material is available online. Information included: design philosophy, 3D printing instructions and workflow, specific instructions for the main components, reactor assembly guides, experimental details, electrochemistry and reference electrodes, future work, 3D-printing terminology, and references. See also doi:10.5281/zenodo.18322571.

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Data availability. The 3D-printed gas diffusion electrode reactor files and firmware are archived on *Zenodo* (doi:10.5281/zenodo.18322571) with source development hosted on GitHub at https://github.com/cdb-usyd/openGDE. The version of the code employed for this study is version 1.0.0.

Conflicts of interest. The authors declare that they have no further conflicts of interest.

Declaration of funding. This work was supported by the Australian Research Council CE230100032 (to A. K. L. Yuen and C. Barnett) and DP220101511 (to A. K. L. Yuen, T. Maschmeyer and C. Barnett).

Acknowledgements. C. Barnett, A. K. L. Yuen and T. Maschmeyer acknowledge the ARC Discovery Project DP220101511 and Sydney Nano. This work was performed on the land of the Gadigal of the Eora Nation. We thank Prof. Jakelin Troy for providing a Gadi language name for this project: bayadjamara, which means 'air maker'. C. Barnett thanks: Dr D. Alam for advice on inlet connections, and Dr T. Hashemzadeh, Dr Y. Y. Cheng, Dr A. Katselas and Dr M. Kang for testing the 3D printing instructions for usability.

Author contributions. T. Maschmeyer initiated the research activity and aided in the discussions around design, fabrication and testing; C. Barnett jointly conceived the idea, designed, coded and printed platform, collected and analysed the data, and prepared the manuscript; A. K. L. Yuen test printed the test-piece and reproduced the reactor assembly; developed, edited and revised the manuscript; A. F. Masters provided experimental discussion and helped edit the manuscript; T. Maschmeyer revised and edited the manuscript over several iterations and helped to prepare for final submission.

Author affiliations

[A]The Laboratory of Advanced Catalysis for Sustainability, The University of Sydney, Sydney, NSW, Australia.

[B]ARC Centre of Excellence for Carbon Science and Innovation at The University of New South Wales, Sydney, NSW, Australia.

[C]Sydney Nano, The University of Sydney, Sydney, NSW, Australia.

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